

HAZOMAT

B39VS - Systems Project

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Abstract

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1 Introduction

1.1 Project overview

The project involves the design and development of an autonomous mobile manipulator capable of picking up waste from waste points and disposing of it at the base station.

The robot, designated ‘Hazmat’ manufactured by SLAM Industries, is equipped with a Raspberry Pi 5 as its main processor, assisted by an ESP32-S3 to offload computational tasks. This configuration reduces the processing burden on the Pi and provides an additional safeguard against potential hardware faults due to faulty wiring.

Hazmat is fitted with mecanum wheels, enabling omni-directional movement and allowing it to smoothly traverse through tight spaces within a factory environment. The robot utilizes a stereo-depth camera (Oak-D Pro W) with an integrated IMU, motor encoders, and a 2D LiDAR (Rplidar A3) to perform simultaneous localization and mapping (SLAM), enabling it to autonomously locate waste containers and navigate to disposal points.

Hazmat is also equipped with a 3-DOF manipulator designed using a four-bar linkage mechanism, and a dual-claw soft servo gripper for efficient waste pickup and placement. The servo motors were modified to output potentiometer feedback to the Pi, enabling real-time inverse kinematics calculations for the robotic arm. Power is supplied by a lithium polymer battery that drives the wheels, while a buck converter is used to step down voltage for the robot’s electronic systems.

1.2 Aims and objectives

Aims:

- To design, fabricate, and demonstrate a functional, autonomous mobile manipulator capable of navigating an indoor environment, picking up hazardous waste containers, and delivering them to designated disposal areas with minimal human intervention.

Objectives:

- Design and fabricate a stable mobile robot based on mecanum wheels to allow omnidirectional movement in constrained indoor environments.
- Develop a secure container compartment to safely transport waste containers during robot motion.
- Integrate vision and ranging sensors (stereo camera, 2D LiDAR, ultrasonic sensors) to enable environmental perception, obstacle detection, and navigation.

- Design and build a 3-DOF robotic manipulator with a four-bar linkage for grasping and moving waste containers.
- Implement autonomous navigation using SLAM (Simultaneous Localization and Mapping) and obstacle avoidance algorithms.
- Implement object and container recognition using colour coding and computer vision methods.
- Develop audio feedback features to alert humans during path obstruction scenarios.
- Integrate micro-ROS middleware to allow reliable communication between the embedded control systems and the central processor.
- Implement a user interface and visual indicators (LEDs or screen) to display robot status and task progress.
- Develop a simple autonomous charging strategy (proposal stage) for future continuous operation.
- Simulate and test robot behaviour in Gazebo or a relevant simulation environment before real-world testing.
- Manage team roles and collaborate effectively across mechanical, electrical, and software domains to achieve system integration.

1.3 Roles and Responsibilities

SLAM Industries started as a military contractor, developing advanced autonomous systems for extreme environments. Our team created robots for reconnaissance, high-risk operations, and more, pushing the limits of robotics technology. Now, we have shifted focus, using our military-grade expertise to design robots for a wide range of industries. From hazardous waste management to autonomous transportation, we deliver reliable, high-performance solutions.

SLAM Industries is a team of six passionate individuals committed to redefining the future of autonomous systems. The company structure is organized as shown in Figure 1, and the key responsibilities and contributions are as follows:

Abdul Maajid Aga – Director of Robotic Operations

- Designed the robot base, top plate, and structural stands, ensuring organized component layout and accessibility.
- Co-developed the 3-DOF four-bar linkage arm with Laith.
- Migrated wheel control from serial to micro-ROS on the ESP32-S3.
- Helped wire and verify the full electrical system.
- Developed inverse kinematics algorithms for the arm (not used in final version).

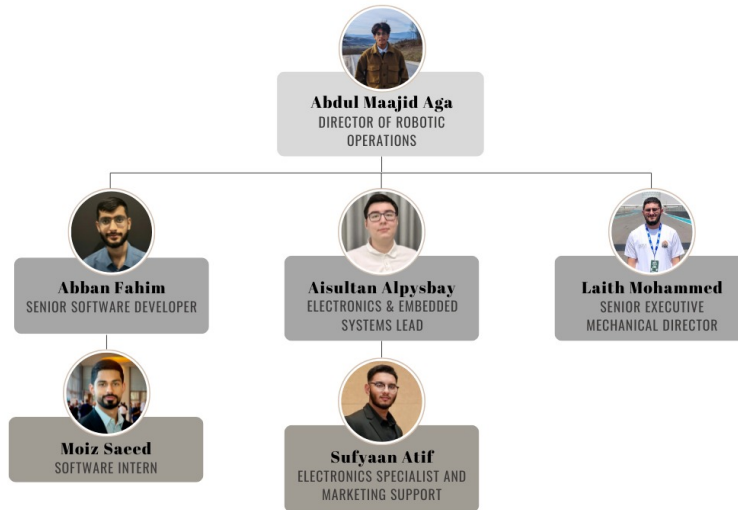


Figure 1: Organisation hierarchy

- Created and refined early prototypes for mechanical testing.
- Optimized the robot's structure for easy assembly, maintenance, and manufacturing.
- 3D-printed and assembled mechanical components.
- Led the team, assigned tasks, enforced timelines, and maintained momentum.
- Provided morale support and ensured teammate well-being throughout the project.

Laith Mohammed – Senior Executive Mechanical Director

- Assisted in designing and prototyping multiple robot base plates using laser-cut acrylic until finalizing the design.
- Designed and prototyped several robotic arm versions using laser-cut acrylic to reach the final version.
- Sourced power converters and built cable harnesses for power distribution.
- Assembled a custom PCB using a perf board to securely integrate the ESP32 with the motor driver board.
- Developed a C++ library to control and read the speed of the mecanum wheels.

- Coordinated between electrical, mechanical, and software teams.

Aisultan Alpysbay – Electronics and Embedded Systems Lead

- Designed the electrical schematic, simplified diagram, and initial PCB (not used in final system).
- Assembled the robotic arm multiple times, including rebuilds after failures.
- Rewired the robot's electronics twice for improved reliability.
- Replaced damaged motor drivers.
- Researched and provided design inspiration for the final robot layout.
- Soldered additional wiring onto four servos for sensor feedback (one later replaced).
- Suggested vision-based alignment techniques to improve arm positioning.

Sufyaan Atif – Electronics Specialist and Marketing Support

- Designed the complete electrical schematic and simplified system power flow diagram.
- Gathered and organized all required electronic components.
- Assembled and wired the robot's electrical system.
- Tested DC motors, encoders, and sensors for functionality.
- Wrote code to read encoder pulses and calculate wheel RPM.
- Calibrated encoders for reliable measurements.
- Assembled the robotic arm and programmed servo operating limits.
- Tested arm movement with different joint angles.
- Managed the project website and marketing activities.

Abban Fahim – Senior Software Developer

- Interfaced sensors with Raspberry Pi, including LiDAR and stereo camera.
- Set up the complete ROS framework, launch scripts, and development environment on Raspberry Pi.
- Implemented inverse kinematics and PID controller for mobile base control.
- Created a simulation environment to test mobile robot behavior.
- Assisted in implementing SLAM and color detection nodes.
- Developed ArUco marker detection and localization system.

- Built the autonomous navigation stack, fusing multi-sensor data and odometry.
- Wrote the initial serial communication prototype between Pi and ESP.
- Programmed final I2C communication system for the manipulator.
- Mentored the team on Linux, version control, and ROS fundamentals.

Moiz Saeed – Software Intern

- Assisted in designing the robot’s base structure.
- Developed and tested the odometry ROS node.
- Supported the implementation of Cartographer SLAM using LiDAR for mapping and localization.
- Wrote the full URDF (Unified Robot Description Format) model for the robot.
- Helped set up the robot’s transformation (TF) frames in ROS.
- Implemented a colour detection system using OpenCV.
- Developed a ROS node for audio feedback and notifications.
- Assisted in the development of the robot’s autonomous navigation system.

1.4 Project scope

This project focused on the design, development, and demonstration of a scaled-down autonomous mobile manipulator intended for hazardous waste handling in an industrial indoor environment. The system combines mechanical design, embedded electronics, and advanced software integration to meet the key operational requirements.

The main elements covered within the project scope included:

- Designed a stable mobile base with mecanum wheels for omnidirectional movement in confined spaces.
- Assembled a 3-DOF manipulator with a four-bar linkage for waste handling.
- Integrated a secure containment compartment to safely transport waste containers.
- Integrated a stereo depth camera and 2D LiDAR for environment perception and mapping.
- Implemented basic colour detection for container identification.
- Developed SLAM-based autonomous navigation between pickup and drop-off points.

- Implemented basic path planning with fallback behaviour for obstacle avoidance.
- Added an audio feedback system to alert humans blocking the path.
- Fully integrated mechanical, electrical, and software subsystems.
- Used micro-ROS for communication between the Raspberry Pi and ESP32 controllers.
- Implemented initial robot status communication via ROS topics.
- Created a simulation environment for validating inverse kinematics, PID control, and navigation algorithms.
- Conducted multiple rounds of physical testing for calibration and optimization.
- Structured task division across team members specializing in mechanics, electronics, software, and marketing.
- Regular design reviews and troubleshooting sessions to address integration challenges.

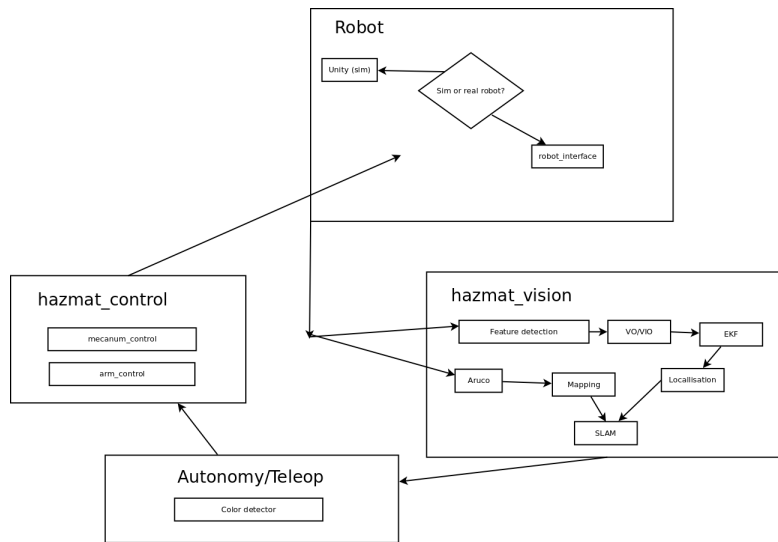


Figure 2: High level overview of the ROS nodes and topics

2 Background and Theory

2.1 Literature review

2.2 Motor Control

2.3 Sensing and Localisation

3 Methodology and Implementation

3.1 Introduction

3.2 Concept Development and Evaluation

3.3 Digital Design

3.4 Mechanical Design

3.4.1 Mobile Base

3.4.2 Manipulator

3.4.3 Assembly

3.5 Sensors

3.6 Software

Our software stack was primarily built upon the expansive ecosystem of Robot Operating System (ROS), which provided cross-script communication, parallelisation, and ready-to-use packages maintained by robotics companies and industry experts, out of the box. Figure 2

- 3.6.1 SLAM
- 3.6.2 Vision
- 3.6.3 Embedded systems
- 3.6.4 Control
- 3.7 Circuit schematics
- 3.8 Prototype development
- 4 Results
 - 4.1 Computer simulations
 - 4.2 Testing results
 - 4.3 Discussions
- 5 Management and Marketing
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- 6 Conclusion and Future Work